

AMEYA SHIRISH BHAVE

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❖ SUMMARY:

PhD researcher and Qiskit Advocate specializing in **quantum error correction (QEC)** with a focus on **Realtime decoding problem**, **erasure-aware fault tolerance** and **scalable decoder + benchmarking pipelines** for QLDPC (BB, HGP) codes. Hands-on experience building **Stim/sinter** simulation workflows, stabilizer-circuit transformations, and **message-passing ML/Classical decoders** with an emphasis on reproducibility and low-latency trade-offs. Contributor to IBM's **Qiskit Advocate Mentorship Program (QAMP)** (C-API circuit library, docs, tutorials). Seeking a **QEC/architecture internship** to build hardware-inspired fault-tolerant tools.

❖ EDUCATION:

The University of Texas at Dallas — **PhD** in Computer Engineering, 2024 - May 2027.

GPA : 3.53/4

University of Maryland, Baltimore County — **MS** in Computer Science, 2022-24.

GPA : 3.63/4

❖ SKILLS:

- **Quantum & QEC:** Qiskit; Stim/sinter; QLDPC codes; Bivariate Bicycle (BB) codes; HGP codes; stabilizer circuits; syndrome-based simulation; erasure-aware workflows; LER benchmarking; pseudo-threshold analysis; decoders (BP, BP+OSD, Relay-BP, ML-based).
- **Fault-Tolerant Systems & Architecture:** Real-time QEC decoding; decoder latency/throughput analysis; queueing and admission control; SRAM-fit and memory-footprint tradeoffs; code-distance/resource scaling; QRE-style modeling; logical reliability analysis.
- **Machine Learning & Data Analysis:** Hypergraph neural networks; GNN-based modeling; PyTorch; training/evaluation pipelines; large-scale simulation-data analysis; performance profiling; visualization and reporting.
- **Programming & Scientific Computing:** Python; Rust; NumPy; SciPy; Pandas; Qiskit; Stim/sinter; PyMatching; Git/GitHub; reproducible benchmarking workflows.
- **Soft Skills:** Communication, Collaboration, Problem Solving, and Technical Writing.
- **Relevant Course Work:** Quantum Information and coding theory, Quantum Communications, Intro to Quantum Computing.

❖ PUBLICATIONS:

- [1] **"BiBiEQ : Bivariate Bicycle Codes on Erasure Qubits"**, *IEEE QCNC 2026*
- [2] **"HyperNQ: A Hypergraph Neural Network Decoder for Quantum LDPC Codes"**, *IEEE ICC 2026*
- [3] **"QUARTET: Quantum Utilization and Adaptation via Resource-Tuned Execution Techniques"**, *2025 IEEE QCE*
- [4] **"ZXNet: ZX Calculus-Driven GNN Framework for Quantum Circuit Equivalence Checking"**, *Proc. 62nd ACM/IEEE DAC, 2025.*

❖ RESEARCH EXPERIENCE:

Research Assistant, UT Dallas, Dallas

, Aug 2024– Present.

1. **Deadline-Driven Framework for Real-Time QLDPC Decoding** – *submitted to IEEE QCE 2026*, Jan – Apr. 2026.
 - A deadline-driven QLDPC decoding framework with **queueing**, **EDF scheduling**, **admission control**, and **bounded rescue** policies.
 - Modeled **SRAM-fit**, **tail-latency**, **overload**, and **capacity-scaling** regimes for BP-style decoders on **BB72/144/288** workloads.
 - Reduced SRAM-fit boundary by **4×** using cached summaries; showed pooled decoder capacity reduced **MissRate 97.64%→0.98%**.
2. **BiBiEQ : Bivariate Bicycle Codes on Erasure Qubits** – *Accepted IEEE QCNC 2026*, May – Oct. 2025.
 - Built **2 conversion engines** (Exact, Approx) to transform **erasure-aware memory circuits (Stim)** into **stabilizer circuits** for decoding.
 - Benchmarked **BB codes 72/108/144**; **LER decreased 26–34×** for **distance (d):6→10** vs **2–2.7×** for **d:10→12** (largest gains by **d≈10**).
 - Produced **per-round LER-erasure** curves and **pseudo-thresholds** in a reproducible pipeline.
3. **HyperNQ: Hypergraph NN Decoder for QLDPC Codes** – *Accepted in IEEE ICC 2026*, Apr – Aug 2025.
 - Designed a **two-stage message-passing HGNN** (hyperedge modeling; linear-in-sparsity) for QEC decoding.
 - Achieved **84% LER decrease** vs BP and **50% vs GNN** in the pseudo-threshold regime with a **shallower network**.
 - Built a reproducible **training/evaluation** stack; profiled **accuracy/throughput trade-offs** for low-latency decoding.

- ❖ **RELEVANT PROJECTS:**
1. **Qiskit Advocate Mentorship Program (QAMP) — Circuit Library (C-API) ·**

, Oct 2025– Jan 2026.

- Built C-API circuit-library scaffold with core and variational circuits, aligned with Qiskit transpiler concepts with documentation.
2. **Solving System of Linear Equations using Quantum Algorithm and finding potential speed-ups**

, Oct - Dec 2023.

- Implemented HHL in **Qiskit** and built a **custom emulator** showing **constant-time per-shot** vs the simulator’s exponential scaling.
 - Benchmarked emulator vs simulator; achieved **~1e-3 absolute error** against ground truth on test systems.
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- ❖ **Technical Certification and Badges:**
- IBM Qiskit Advocate Mentorship Program 2025 Contributor

, Jan 2026.
- IBM Quantum Machine Learning

, Dec 2025.
- IBM Quantum Chemistry, with Variational Quantum Eigensolver

, Dec 2025.
- IBM Variational Algorithm Design

, Dec 2025.
- IBM Certified Associate Developer - Quantum Computation using Qiskit v0.2X

, Nov 2021.